

# Chapter 3-4

## Multiplexing

# Objectives

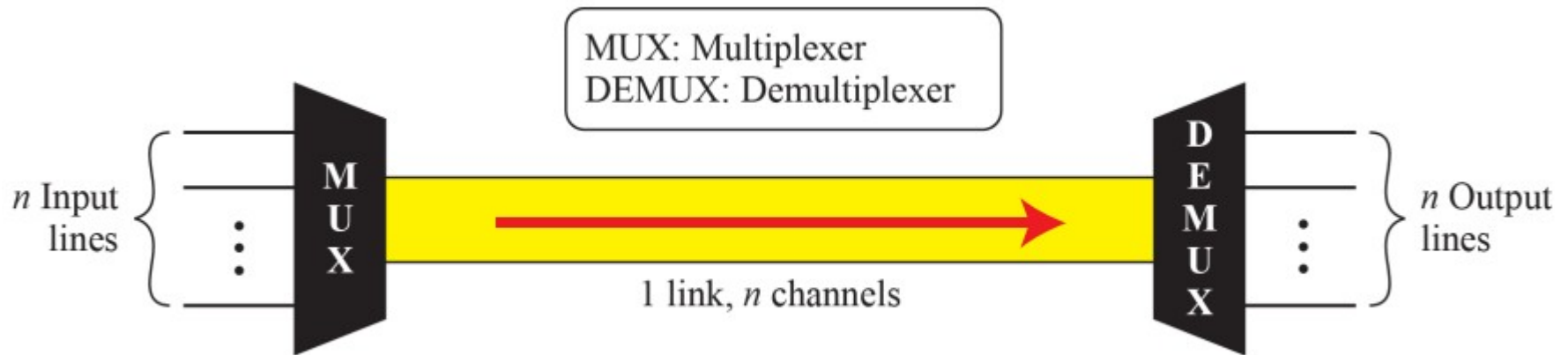
- Describe frequency division multiplexing and list its applications, advantages, and disadvantages .
- Describe synchronous time division multiplexing and list its applications, advantages, and disadvantages
- Outline the basic multiplexing characteristics of both T1 and ISDN telephone systems
- Describe statistical time division multiplexing and list its applications, advantages, and disadvantages
- Cite the main characteristics of dense wavelength division multiplexing and its advantages and disadvantages
- Cite the main characteristics of code division multiplexing and its advantages and disadvantages

# Multiplexing

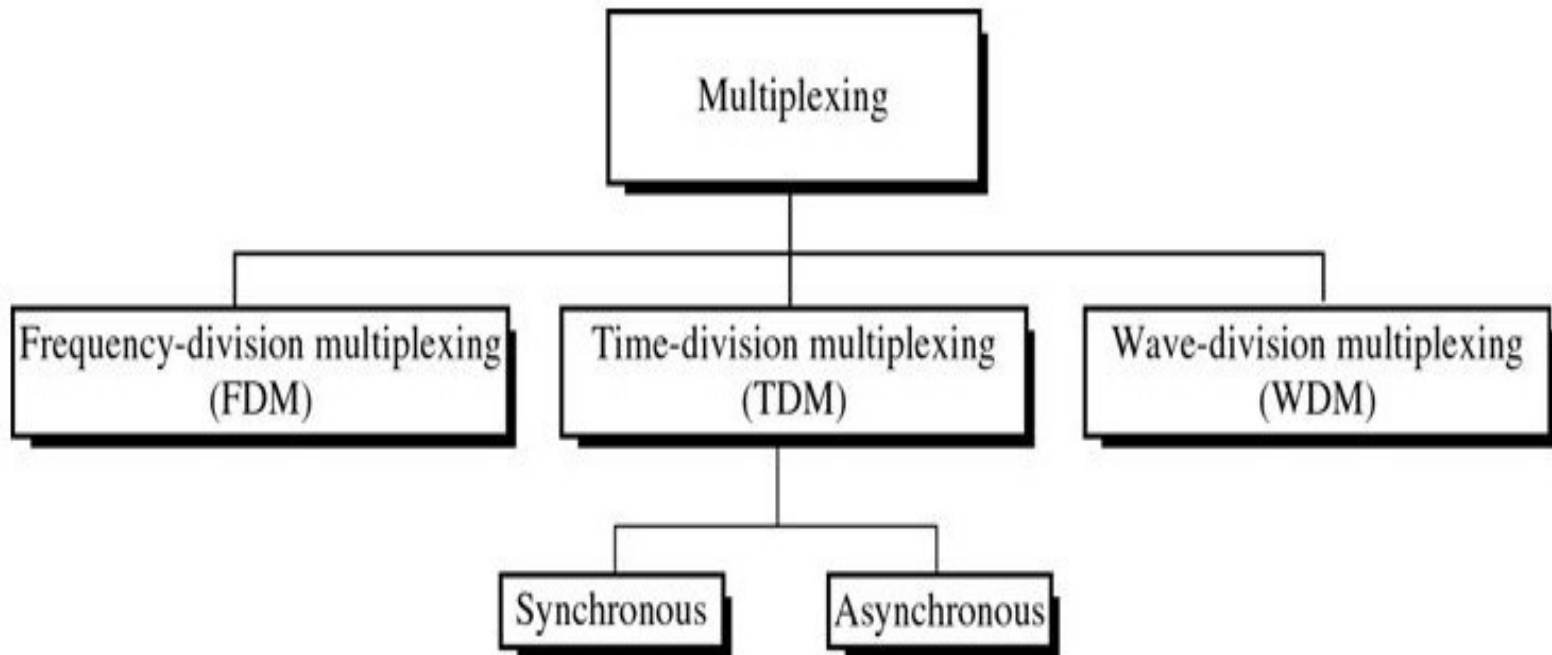
- Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared.
- Multiplexing is the set of techniques that allows the simultaneous transmission of **multiple signals** across a **single data link**.
- Multiplexing is a technique performed at the physical layer of the OSI model or the interface layer of the Internet model.

# Dividing the link in to channels

- In a multiplexed system,  $n$  lines share the bandwidth of one link.



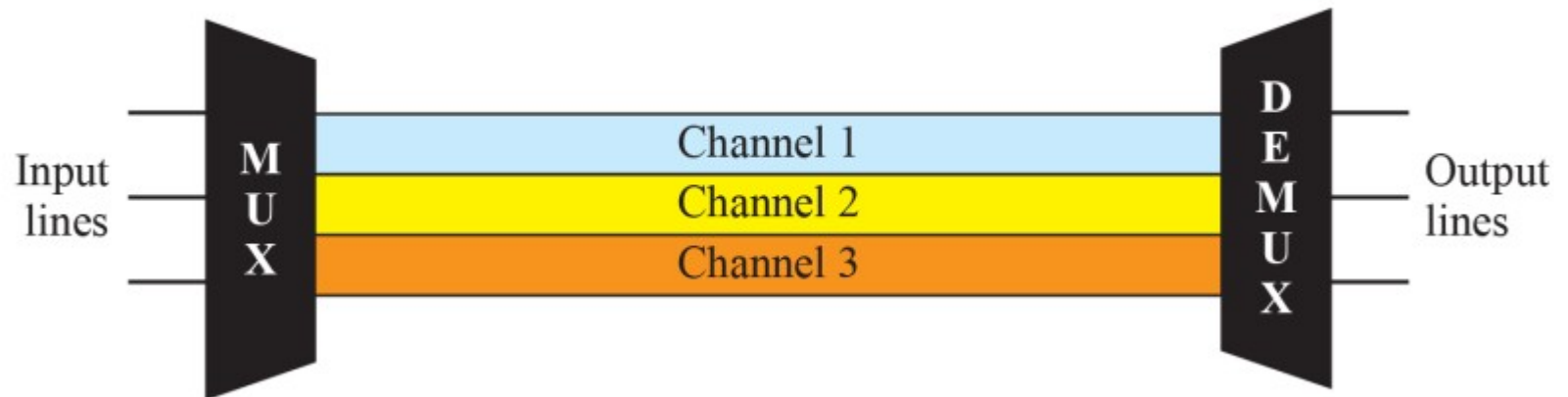
# Categories of Multiplexing

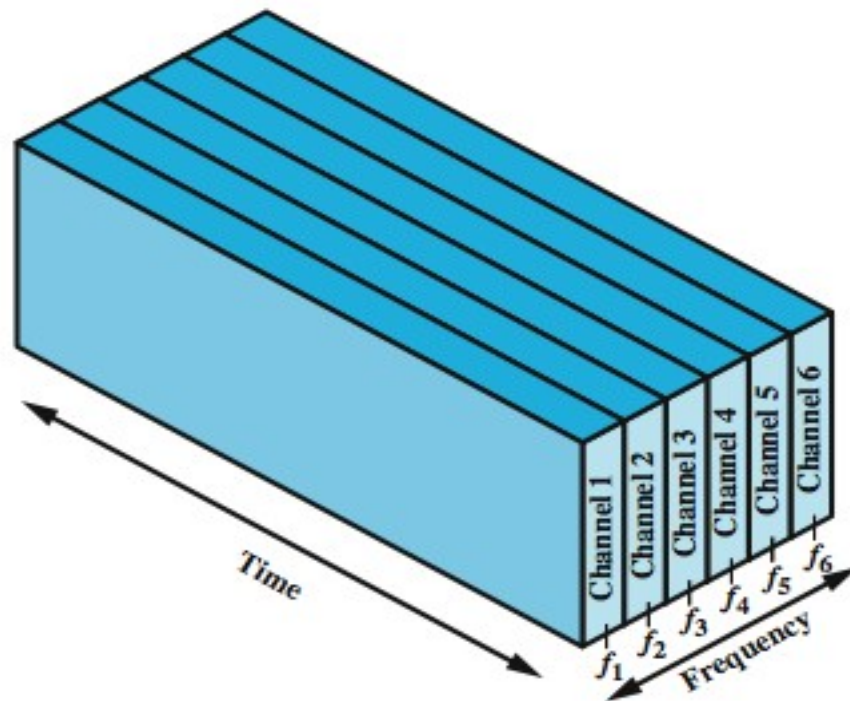


- The first two techniques are designed for analog signals, the third, for digital signals

# Frequency Division Multiplexing(FDM)

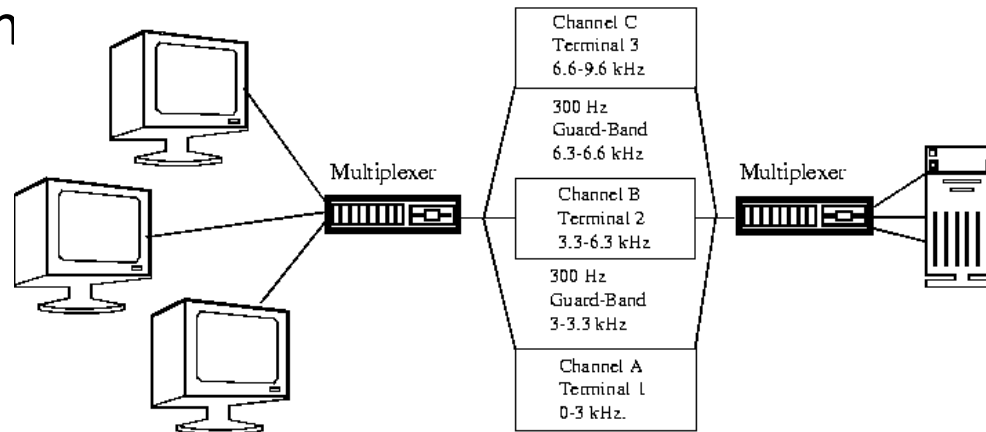
is an analog technique where each communications channel is assigned a carrier frequency.





(a) Frequency division multiplexing

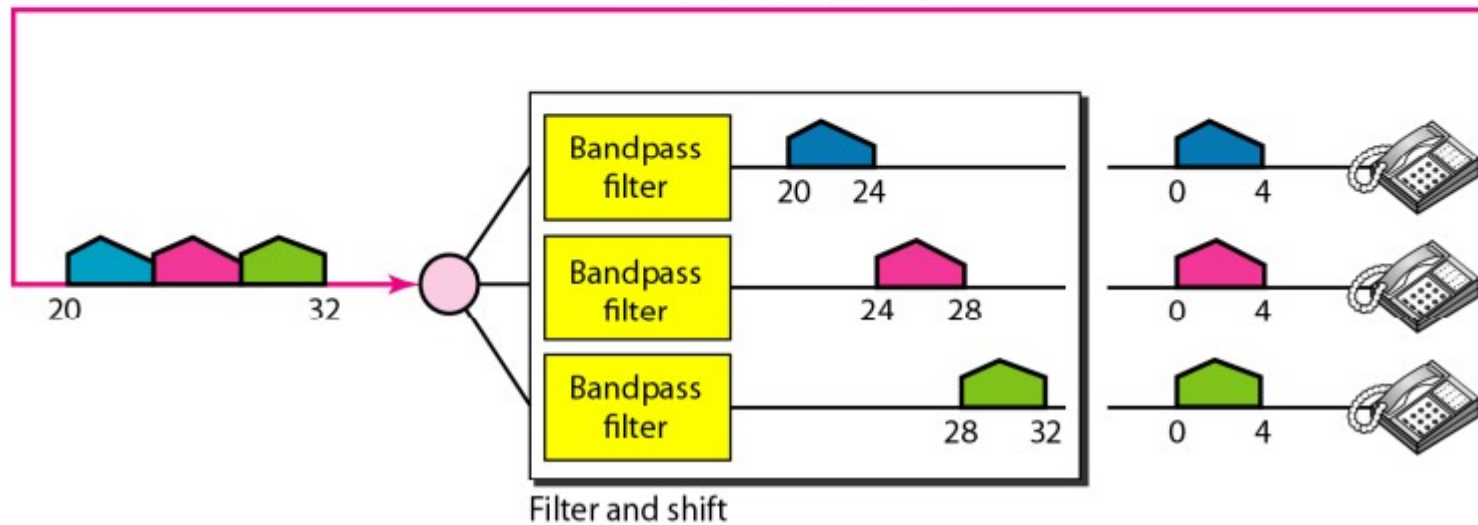
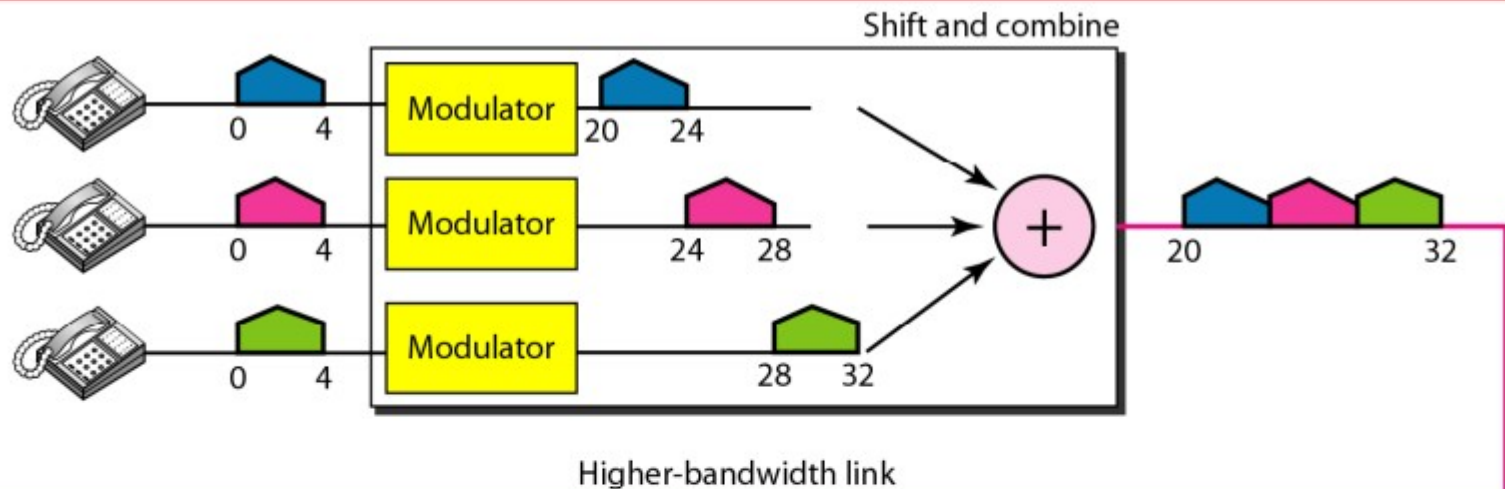
- can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted .
- To separate the channels, a **guard-band** would be used. This is to ensure that the channels do not interfere with each other
- Used in



# Example

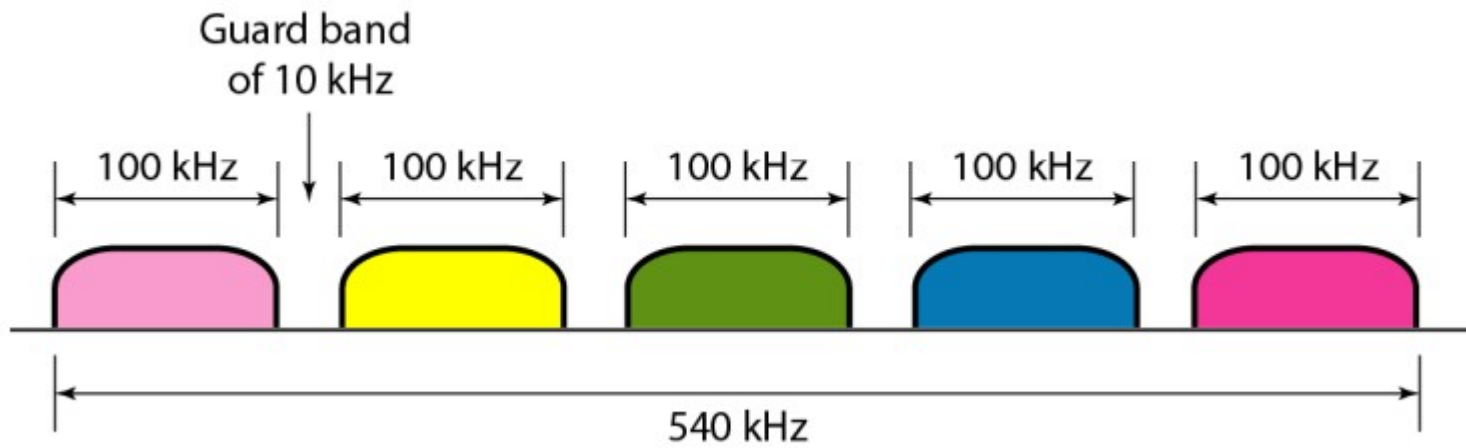
- Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

- Solution
- We shift (modulate) each of the three voice channels to a different bandwidth as shown in Figure different bandwidth, as shown in Figure . We use the We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel and the kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them as shown in Figure 6 6 6.10 them as shown in Figure .



# Example

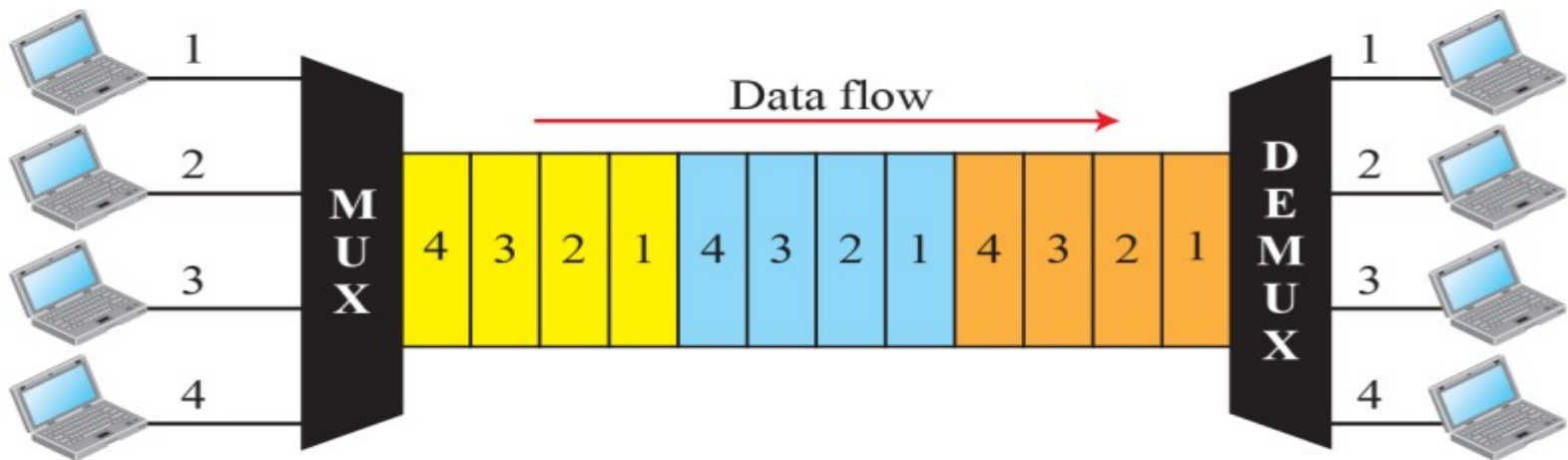
- Five channels channels, each with a 100-kHz bandwidth bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?
- Solution
- For five channels, we need at least four guard bands. This means that the required bandwidth is at least  $5 \times 100 + 4 \times 10 = 540$  kHz.

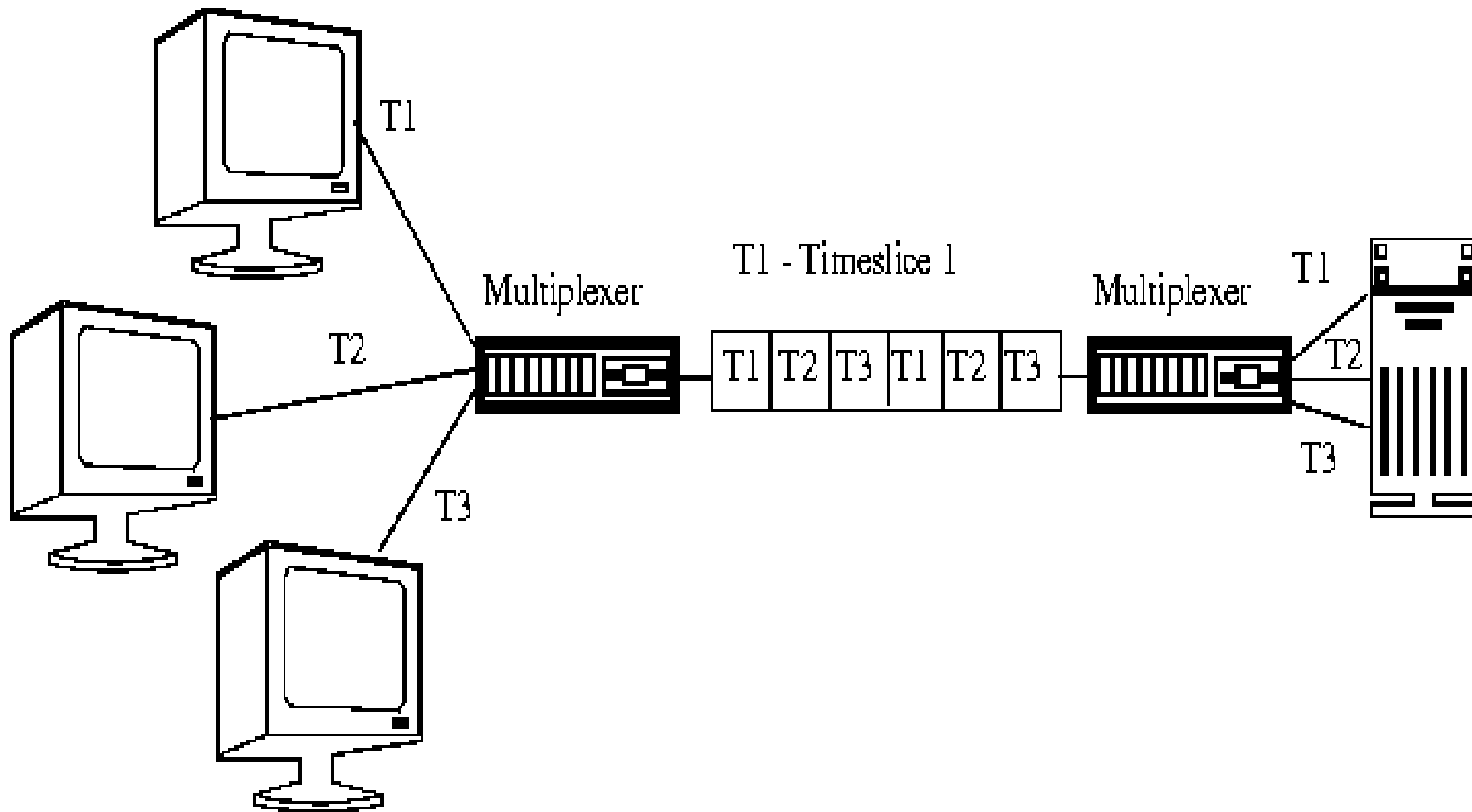


# Time division Multiplexing(TDM)

- allows several connections to share the high bandwidth of a link.
- instead of sharing a portion of the bandwidth as in FDM, time is shared.
- is a technique where a short time sample of each channel is inserted into the multiplexed data stream. Each channel is sampled in turn and then the sequence is repeated.
- each connection occupies a portion of time in the link.
- in the figure, portions of signals 1, 2, 3, and 4 occupy the link sequentially.

- TDM is more efficient, easier to operate, less complex and less expensive compare to FDM.
- TDM which is better suited for digital data



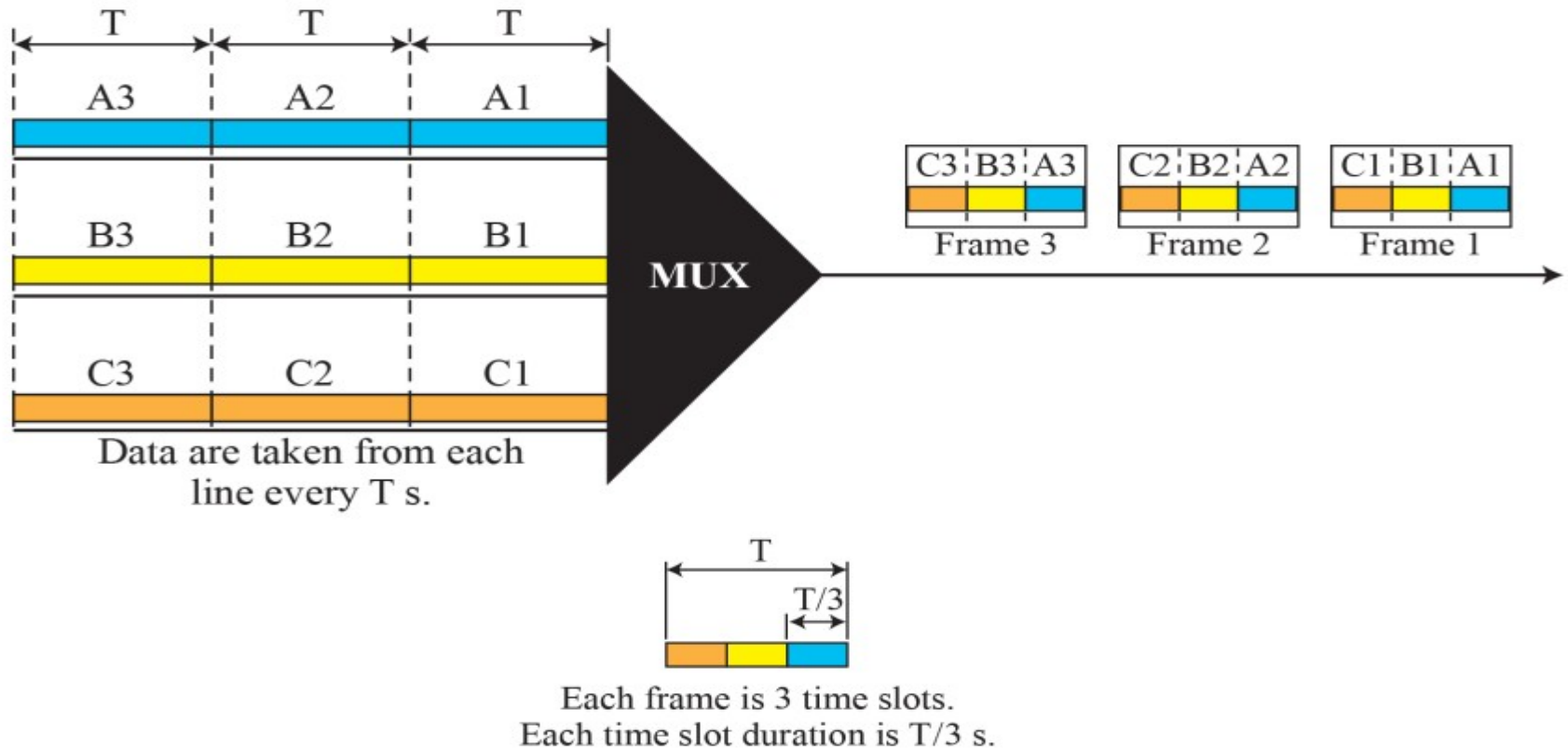


- TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.

# Synchronous TDM

- Is simple time division multiplexing, each conversation is given one slot per frame
  - Guaranteed the slot
  - Slot is wasted if the connection does not use it
  - Wasteful but still brings economies of scale
  - Slot not Used In expensive to implement
- In synchronous TDM, the data rate of the link is  $n$  times faster, and the unit duration is  $n$  times shorter.

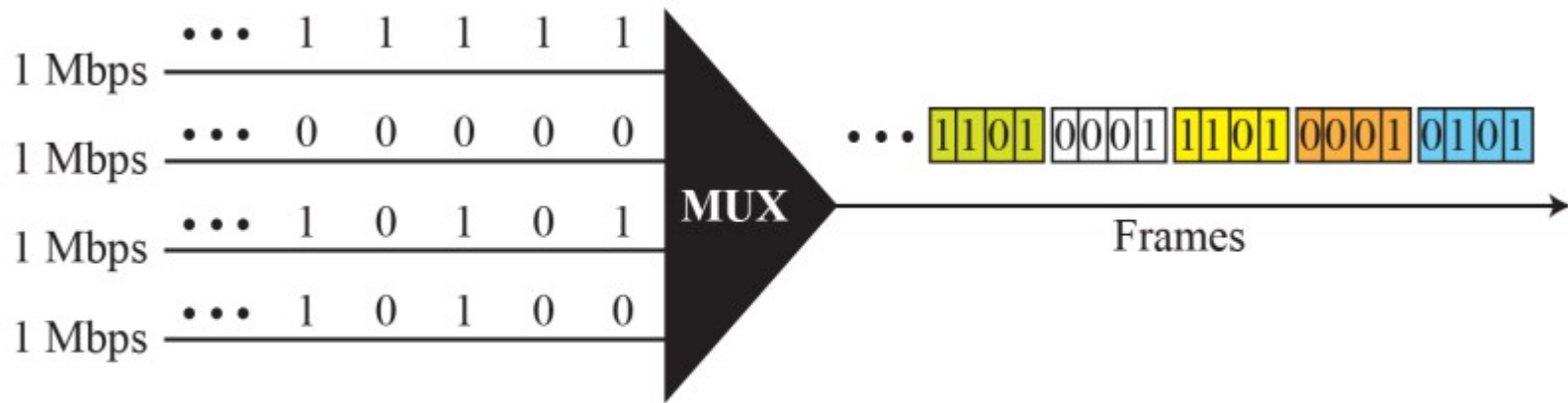
# Synchronous time-division multiplexing



- In the previous picture, the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of
  - each input slot,
  - each output slot, and
  - each frame?

- **Solution**

- 1. The data rate of each input connection is 1 kbps. This means that the bit duration is  $1/1000$  s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).
- 2. The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is  $1/3$  ms.
- 3. Each frame carries three output time slots. So the duration of a frame is  $3 \times (1/3)$  ms, or 1 ms. The duration of a frame is the same as the duration of an input unit.



Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (1) the duration of 1 bit before multiplexing, (2) the transmission rate of the link, (3) the duration of a time slot, and (4) the duration of a frame.

### Solution

We can answer the questions as follows:

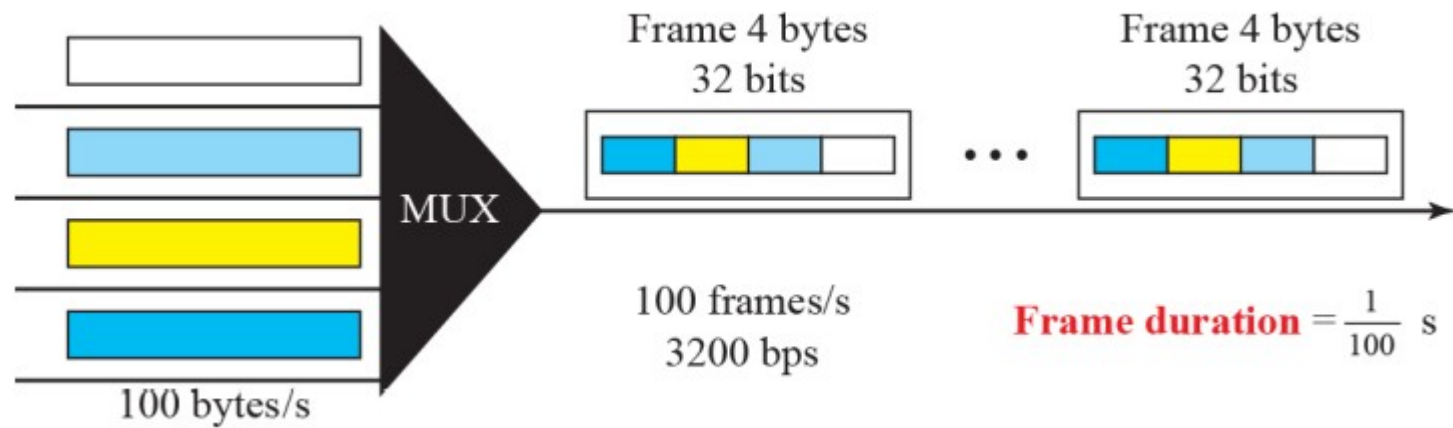
1. The duration of 1 bit before multiplexing is  $1/1$  kbps, or 0.001 s (1 ms).
2. The rate of the link is 4 times the rate of a connection, or 4 kbps.
3. The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or  $1/4$  ms or 250  $\mu$ s. Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or  $1/4$  kbps or 250  $\mu$ s.
4. The duration of a frame is always the same as the duration of a unit before multiplexing, or 1 ms. We can also calculate this in another way. Each frame in this case has four time slots. So the duration of a frame is 4 times 250  $\mu$ s, or 1 ms.

Four channels are multiplexed using TDM. If each channel sends 100 bytes/s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.

### **Solution**

The multiplexer is shown in Figure on next page. Each frame carries 1 byte from each channel; the size of each frame, therefore, is 4 bytes, or 32 bits. The frame rate is 100 frames per second. The duration of a frame is therefore  $1/100$  s. The link is carrying 100 frames per second, and since each frame contains 32 bits, the bit rate is  $100 \times 32$ , or 3200 bps.

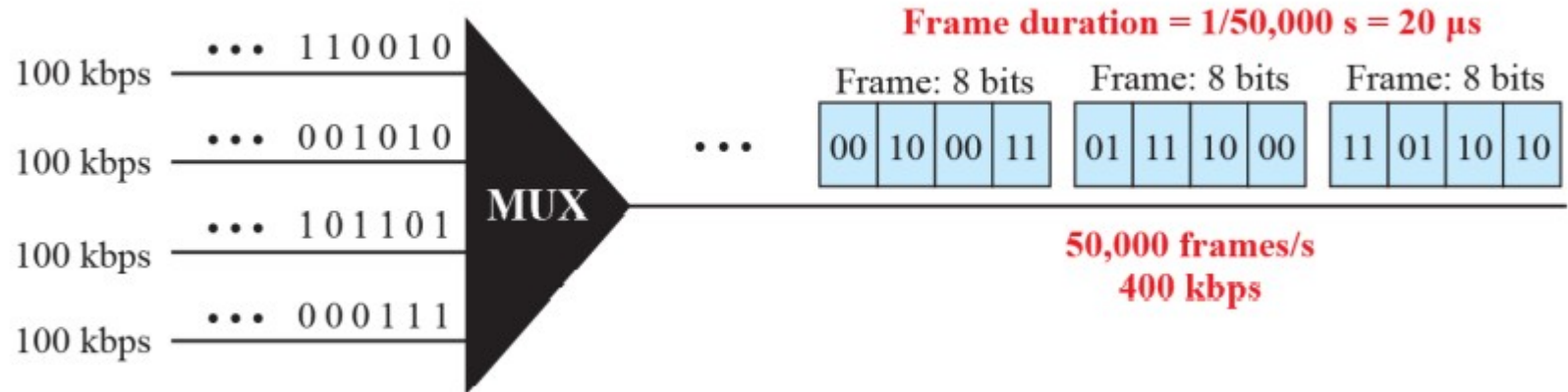
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A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?

### **Solution**

Figure on next page shows the output for four arbitrary inputs. The link carries 50,000 frames per second since each frame contains 2 bits per channel. The frame duration is therefore  $1/50,000$  s or 20  $\mu$ s. The frame rate is 50,000 frames per second, and each frame carries 8 bits; the bit rate is  $50,000 \times 8 = 400,000$  bits or 400 kbps. The bit duration is  $1/400,000$  s, or 2.5  $\mu$ s.



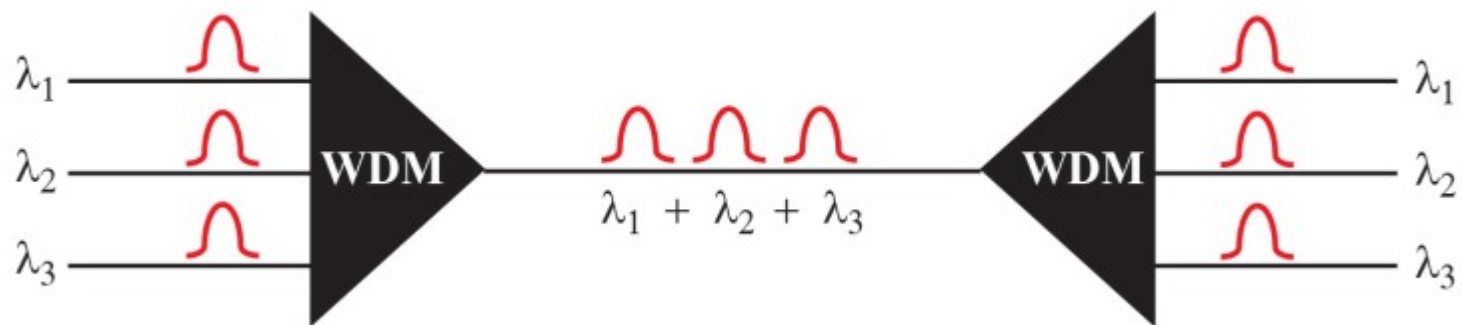
# Asynchronous TDM

- Still Frames and Slots
- but statistical TDM assigns slots as needed
  - Connections that need more slots get them
  - More efficient use of line than simple TDM
  - More expensive to implement Frame

# Wavelength Division Multiplexing

- Dense wavelength division multiplexing is a fairly new technology that multiplexes multiple data streams onto a single fiber optic line.
- It is useful to increase the information carried by single optical fiber.
- WDM can be view as an optical domain version of FDM in which multiple information signal modulate optical signals at different optical wavelength (colors). The resulting signals are combine and transmitted simultaneously over the same optical fiber.

- Unlike frequency division multiplexing, which assigns input sources to separate sets of frequencies, and time division multiplexing, which divides input sources by time, wave division multiplexing uses different **wavelength lasers** to transmit multiple signals.



# *Code Division Multiplexing*

- This multiplexing technique assigns a multiple-bit sequence to each transmitter's binary 1 and binary 0.
- When all transmitters transmit at the same time, the receiver sums all received values and performs a mathematical computation on the sums to separate one code from another.

# *Comparison of Multiplexing Techniques*

- Frequency division multiplexing relies on analog signaling and is the simplest and most noisy of all the multiplexing techniques.
- Synchronous time division multiplexing is also relatively straight forward, and like frequency division multiplexing, input devices that have nothing to transmit can waste transmission space. The big advantage of synchronous TDM is the lower noise during transmission.
- Statistical TDM is one step above synchronous TDM because it transmits data only from those input devices that have data to transmit. Thus, statistical TDM wastes less bandwidth on the transmission link.
- Dense wavelength division multiplexing is a very good, albeit expensive, technique for transmitting multiple concurrent signals over a fiber optic line.